

WEEKLY PROJECT PROGRESS REPORT (WPPR)–

For week commencing 7 March 2026

Programme: BCA (Artificial intelligence, Cloud Computing & Devops)

Student Name: Mohamed rifan p

WPPR: WEEK 2 Register Number: 23BCAICD052 Internal Guide's Name: MISS Niharika

MAJOR PROJECT Title:

Cloud Cost Intelligence for Retail

Targets set for the week:

The targets for week two were to create the Streamlit application with basic layout (sidebar, main area, and five tabs), implement CSV file upload with <code>@st.cache_data</code> caching, add sidebar filters for department, service type, region, and date range, develop the Budget Guard agent to project month-end spend and compare it with a user-defined budget, display key performance indicators (total cost, total units, average unit cost, top department, and month-over-month change), and test the system with a 5,000-row synthetic dataset.

Progress/Achievements for the week:

All targets were met successfully. A Streamlit app skeleton was created with five tabs named Overview, Agents, Insights, Optimization, and Data. The file uploader was implemented and combined with <code>@st.cache_data</code> , which significantly reduced reload times. Dynamic sidebar filters using <code>multiselect</code> and <code>date_input</code> were added, allowing users to slice data by department, service type, region, and date range. The Budget Guard agent was fully developed, including the <code>month_end_projection()</code> function and a budget comparison feature that outputs severity levels (High/Medium/Low). Five KPI cards were displayed at the top of the Overview tab. The system processed 5,000 rows in under one second, meeting performance expectations.

Future Work Plans:

Next week’s focus will be on implementing the Anomaly Investigator agent using a rolling Z-score with a 14-day window, developing the Spend Driver agent for month-over-month variance analysis by service type and department, and adding basic charts (monthly spend line chart and cost-by-service bar chart) to the Overview tab.

Implementa on shown: No

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Yes

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Remarks by the Internal Guide:

Signature of the student

Signature of the Internal Guide